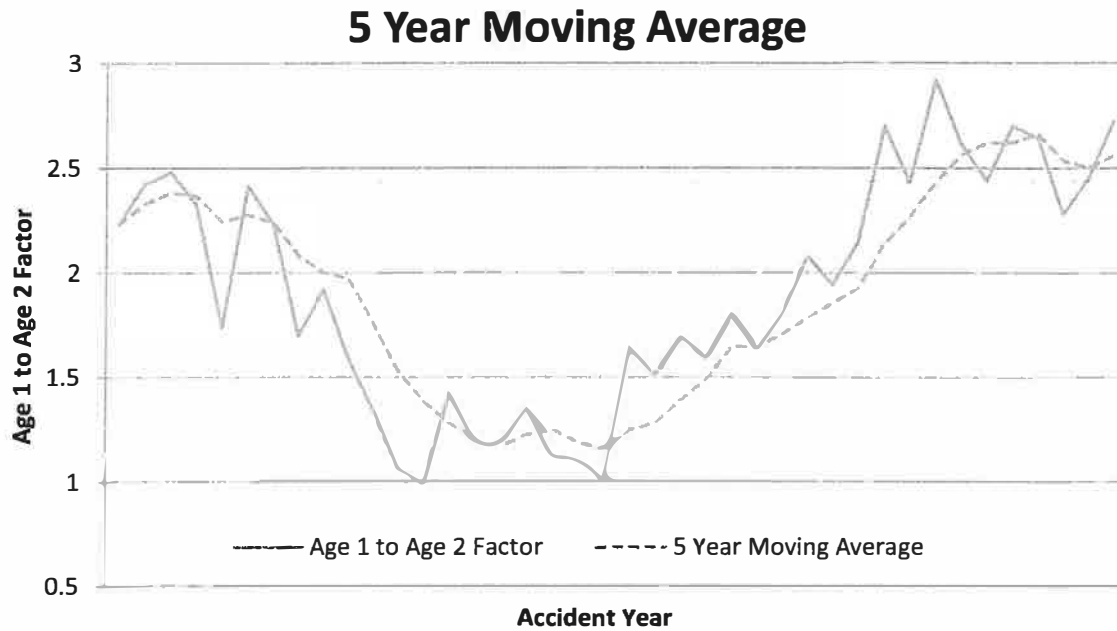


9. (1.25 points)

Given the following information:



- The all-year weighted average is used to calculate the age 1 to age 2 factor.
- An actuary believes that the all-year weighted average is inappropriate.

a. (0.75 point)

Support the actuary's claim and provide two methods that could improve the estimated age-to-age factor considering the chart above.

b. (0.5 point)

Describe an alternative method of testing the stability of the age-to-age factors based on Venter Factors.

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EXAM 7 SPRING 2018 SAMPLE ANSWERS AND EXAMINER'S REPORT

SPRING 2018 EXAM 7, QUESTION 9	
TOTAL POINT VALUE: 1.25	LEARNING OBJECTIVE(S): A2, A3
SAMPLE ANSWERS	
Part a: 0.75 point	
<u>Sample responses for supporting the actuary's claim that the all-year weighted average is inappropriate</u> • The age to age factor is not stable over time. • The 5-year moving average is not stable. • The age to age factor shows an increasing trend. • There has been a significant change in the age to age factor over time. • The age to age factor has been higher in recent years. • There is a clear shift in the graph. • There is a change in the level around which the age to age factor is fluctuating. <u>Sample responses for providing two methods that could improve the estimated age to age factor</u> • Use a weighted average with more weight on recent years. • Use a 5-year weighted average. • Use the 5-year moving average. • Use a 5-year simple average. • Use a 5-year ex hi/lo average. • Group accident years. • Exclude the accident years where the age to age factors are lower. • Use the state-space model • Fit a curve to age to age factors. • Use a time series analysis. • Use expert opinion to select the age to age factor. • Use industry data to select the age to age factor. • Adjust the data for systematic changes. • Use Berquist-Sherman method to adjust data. • Use a Bayesian method which incorporates expert opinion. • Use ODP GLM on the latest 5 diagonals. • Use CCL model. • Use CSR model. • Compare paid age to age factors to reported age to age factors.	
Part b: 0.5 point	
• Plot residuals against time. If there are strings of positives and negatives then development factors are not stable. • Use the state-space model which compares the volatility of observations around the mean to the volatility in the mean itself over time. • Look for changes in the settlement rate of claims by measuring the percentage of closed claims closed by age.	

EXAM 7 SPRING 2018 SAMPLE ANSWERS AND EXAMINER'S REPORT

EXAMINER'S REPORT

Candidates were expected to have knowledge of the six testable implications of the assumptions needed for the chain ladder method to be optimal. This question addressed the fourth testable implication discussed in the Venter text: stability of loss development factors. Candidates were expected to recognize instability by analyzing the graph provided, suggest methods that can be used when instability is present, and discuss a method of testing for instability.

Part a

Candidates were expected to identify the instability in the age to age factor and provide two methods that could be an improvement over the all-year weighted average given the trend in recent accident years. Responses that only indicated volatility in the age to age factor did not receive credit as random fluctuation could actually be an argument for using the all-year weighted average.

Common mistakes included:

- Listing only one method to improve the estimated age to age factor.
- Providing support for using the all-year weighted average.
- Listing two alternate methods without providing any support for the actuary's claim.

Part b

Candidates were expected to describe a method for testing stability discussed in the Venter text other than looking for trends in the moving average.

Common mistakes included:

- Describing other testable implications discussed in the Venter text such as significance or linearity.